

$$x^\mu = \Lambda^\mu_\nu x^\nu, \quad x^2 = x^\mu x_\mu = g_{\mu\nu} x^\mu x^\nu = \vec{x}^2 - c^2 t^2$$

$$\therefore g_{\mu\nu} = \Lambda^\sigma_\mu g_{\sigma\lambda} \Lambda^\lambda_\nu, \quad g_{\mu\nu} = \eta_{\mu\nu} = (-, +, +, +)$$

$$\det(\eta_{\mu\nu}) = -1 \Rightarrow \det(\Lambda) = \pm 1$$

$$-1 = \Lambda^\sigma_0 g_{\sigma\lambda} \Lambda^\lambda_0 \Rightarrow (\Lambda^\sigma_0)^2 = 1$$

$$\Lambda^\sigma_0 = \pm 1 \Rightarrow \Lambda^\sigma_0 = 1$$

$$\Lambda^\sigma_i = \Lambda^i_0 = 0 \quad | \quad \Lambda^i_j = R_{ij} \in SO(4)$$

$$\Lambda^\rho_\nu \Lambda^\nu_\sigma = \delta^\rho_\sigma$$

$$\bullet \text{ Inf. Lorentz: } \Lambda^\mu_\nu = \delta^\mu_\nu + \delta\omega^\mu_\nu$$

$$g_{\mu\nu} = (\delta^\sigma_\mu + \omega^\sigma_\mu)(\delta^\lambda_\nu + \omega^\lambda_\nu) g_{\sigma\lambda}$$

$$g_{\mu\nu} = (\delta^\sigma_\mu \delta^\lambda_\nu + \omega^\sigma_\mu \delta^\lambda_\nu + \delta^\sigma_\mu \omega^\lambda_\nu + \omega^\sigma_\mu \omega^\lambda_\nu) g_{\sigma\lambda}$$

$$= g_{\mu\nu} + \omega_{\mu\nu} + \omega_{\nu\mu} + (\omega)^2$$

$$\therefore \boxed{\omega_{\mu\nu} = -\omega_{\nu\mu}}$$

$$\delta\omega_{ij} = -\varepsilon_{ijk} \hat{n}_k \delta\theta \quad (\text{Rotations})$$

$\leadsto \delta\theta$: Rotation by an angle $\delta\theta$ about the unit vector $\hat{n}_k$

$$\delta\omega_{i0} = \hat{n}_i \delta\beta \quad (\text{Boosts})$$

$$\text{Orthochronous Lorentz: } \Lambda^0_0 \geq 1$$

$\bullet$ Lorentz trans. that can be reached by compounding infinitesimal ones are proper
 $\det(\Lambda) = +1$ (Proper)

$$O^+(3,1) \xrightarrow{\text{Parity}} \bigcirc \leftarrow \begin{matrix} \text{improper} \\ \text{or} \text{ } \end{matrix}$$

time reversal
 improper
 nonorthochronous

Parity: $P^\mu_\nu = (P^{-1})^\mu_\nu = \begin{pmatrix} +1 & & \\ & -1 & \\ & & -1 \end{pmatrix}$

* Symmetries: Unitary operators associated to Λ

$U(\Lambda)$: Representation of $O^\uparrow(3,1)$ which is a homomorphism.

$$U(\Lambda\Lambda') = U(\Lambda)U(\Lambda')$$

$$U(\Lambda) = U(\mathbb{1} + \delta\omega) = U(\mathbb{1}) + \delta\omega_{\mu\nu} \frac{\delta U^{\mu\nu}}{\delta\omega} + \mathcal{O}(\delta\omega^2)$$

since U is a rep $U: e \mapsto e_V$ so

$$U(\mathbb{1}) = \mathbb{1}$$

Here

$$\frac{\delta U^{\mu\nu}}{\delta\omega} = \frac{i}{2\hbar} M^{\mu\nu}$$

(Generators of the L.A.)
 Since $e^{i\lambda T} = \mathbb{1} + i\lambda T$

$M^{\mu\nu}$: Generators of the Lorentz group.

* Start with

$$U(\Lambda^{-1})U(\Lambda')U(\Lambda) = U(\Lambda^{-1}\Lambda'\Lambda)$$

$$U(\Lambda^{-1}) = U(\Lambda)^{-1} \text{ and let } \Lambda' = \mathbb{1} + \delta\omega'$$

$$\begin{aligned} U(\Lambda)^{-1}(\mathbb{1} + \delta\omega^{\mu\nu} M_{\mu\nu})U(\Lambda) &= U(\Lambda^{-1}(\mathbb{1} + \delta\omega')\Lambda) \\ U(\Lambda)^{-1}U(\Lambda) &+ U(\Lambda)^{-1}\delta\omega^{\mu\nu} M_{\mu\nu}U(\Lambda) \\ &= U(\Lambda^{-1}\Lambda) + (\Lambda^{-1}\delta\omega'\Lambda) \frac{\delta U}{\delta\omega} \end{aligned}$$

$$\therefore U(\Lambda)^{-1}\delta\omega'^{\mu\nu} M'_{\mu\nu}U(\Lambda) = (\Lambda^{-1})^\mu_\sigma \delta\omega'^{\nu\rho} \Lambda^\sigma_\lambda M^\rho_\lambda$$

* Thus, we find

$$U(\Lambda)^{-1} M^{\mu\nu} U(\Lambda) = \Lambda^\mu_\rho \Lambda^\nu_\sigma M^{\rho\sigma}$$

Any operator carrying one or more vector indices should behave similarly

$$U(\Lambda)^{-1} P^\mu U(\Lambda) = \Lambda^\mu_\nu P^\nu$$

* By expanding $\Lambda = 1 + \delta\omega$ in $\textcircled{A}$ we find

$$[M^{\mu\nu}, M^{\alpha\lambda}] = i\hbar (g^{\mu\rho} M^{\nu\sigma} - (\mu \leftrightarrow \nu)) - (\rho \leftrightarrow \sigma)$$

$$J_i \equiv \frac{1}{2} \epsilon_{ijk} M^{jk} \quad (\text{Rotations})$$

$$K_i \equiv M^{i0} \quad (\text{Boosts})$$

Time evolution in the Heisenberg picture:

$$e^{iHt/\hbar} \psi(\vec{x}, 0) e^{-iHt/\hbar} = \psi(\vec{x}, t)$$

which needs a relativistic generalization

• Unitary spacetime translation operator

$$T(a) \equiv \exp[-i P^\mu a_\mu / \hbar]$$

$$T(a)^{-1} \phi(x) T(a) = e^{i P^\mu a_\mu} \phi(x) e^{-i P^\mu a_\mu}$$

$$= (1 + i) \phi (1 - i) |0\rangle$$

$$= (1 + i) (\phi - \phi i) |0\rangle$$

$$= (\phi + i\phi - \phi i) |0\rangle$$

$$= (\phi + [P, \phi]) |0\rangle$$

$$E \phi = \dots \quad T P \phi T^{-1} = (P \phi)$$

$\therefore$

$$T(a)^{-1} \varphi(x) T(a) = \varphi(x) + q_m(p_m \varphi(x))$$

but $p_m \sim \partial_m$ so

RHS:

$$T(a)^{-1} \varphi(x) T(a) = \varphi(x - a)$$

$$U(\Lambda)^{-1} \varphi(x) U(\Lambda) = \varphi(\Lambda^{-1}x)$$



Claim: $U(\Lambda) |k_1 \dots k_n\rangle = |\Lambda k_1, \dots \Lambda k_n\rangle$

proof:

$$U(\Lambda)^{-1} \varphi(x) U(\Lambda) = \varphi(\Lambda^{-1}x)$$

$$\therefore U(\Lambda)^{-1} \left(\int \frac{d^3k}{(2\pi)^3 \sqrt{2\omega_k}} a_k e^{ikx} + a_k^\dagger e^{-ikx} \right) U(\Lambda)$$

$$U(\Lambda)^{-1} a_k U(\Lambda)$$

RHS: $\int \frac{d^3p}{(2\pi)^3 \sqrt{2\omega_p}} a(p) e^{ip(\Lambda^{-1}x)} + \dots$

$$p^\mu (\Lambda_m^{\nu-1}) x_\nu \Rightarrow k^\mu \equiv p^\sigma (\Lambda_\sigma^\mu)^{-1}$$

$$d^3p = dp^1 dp^2 dp^3 = \int d^3k$$

Since $\det(\Lambda) = 1$

and $|p|$ is invariant as well, so

$$\therefore \underline{p^\mu = \Lambda_\sigma^\mu k^\sigma}$$

$$\int \frac{d^3k}{(2\pi)^3 \sqrt{2\omega_k}} a(k) e^{ikx}$$

and so we conclude

$$U(\Lambda)^{-1} a(k) U(\Lambda) = a(\Lambda^{-1}k)$$

*

$U(\Lambda) |k_1 \dots k_n\rangle = U(\Lambda) a_1^\dagger \dots a_n^\dagger |0\rangle$
 insert $U(\Lambda^{-1})U(\Lambda)$ between each and at the beginning

$$\left[U(\Lambda) a_1^\dagger U(\Lambda^{-1}) \right] \left[U(\Lambda) a_2^\dagger U(\Lambda^{-1}) \right] \dots \left[U(\Lambda) a_n^\dagger U(\Lambda^{-1}) \right] U(\Lambda) |0\rangle$$

$$a_1^\dagger(\Lambda k_1) a_2^\dagger(\Lambda k_2) \dots a_n^\dagger(\Lambda k_n) U(\Lambda) |0\rangle$$

But

$$U(\Lambda) |0\rangle = |0\rangle \quad \text{so}$$

$$U(\Lambda) |k_1 \dots k_n\rangle = |\Lambda k_1 \dots \Lambda k_n\rangle$$